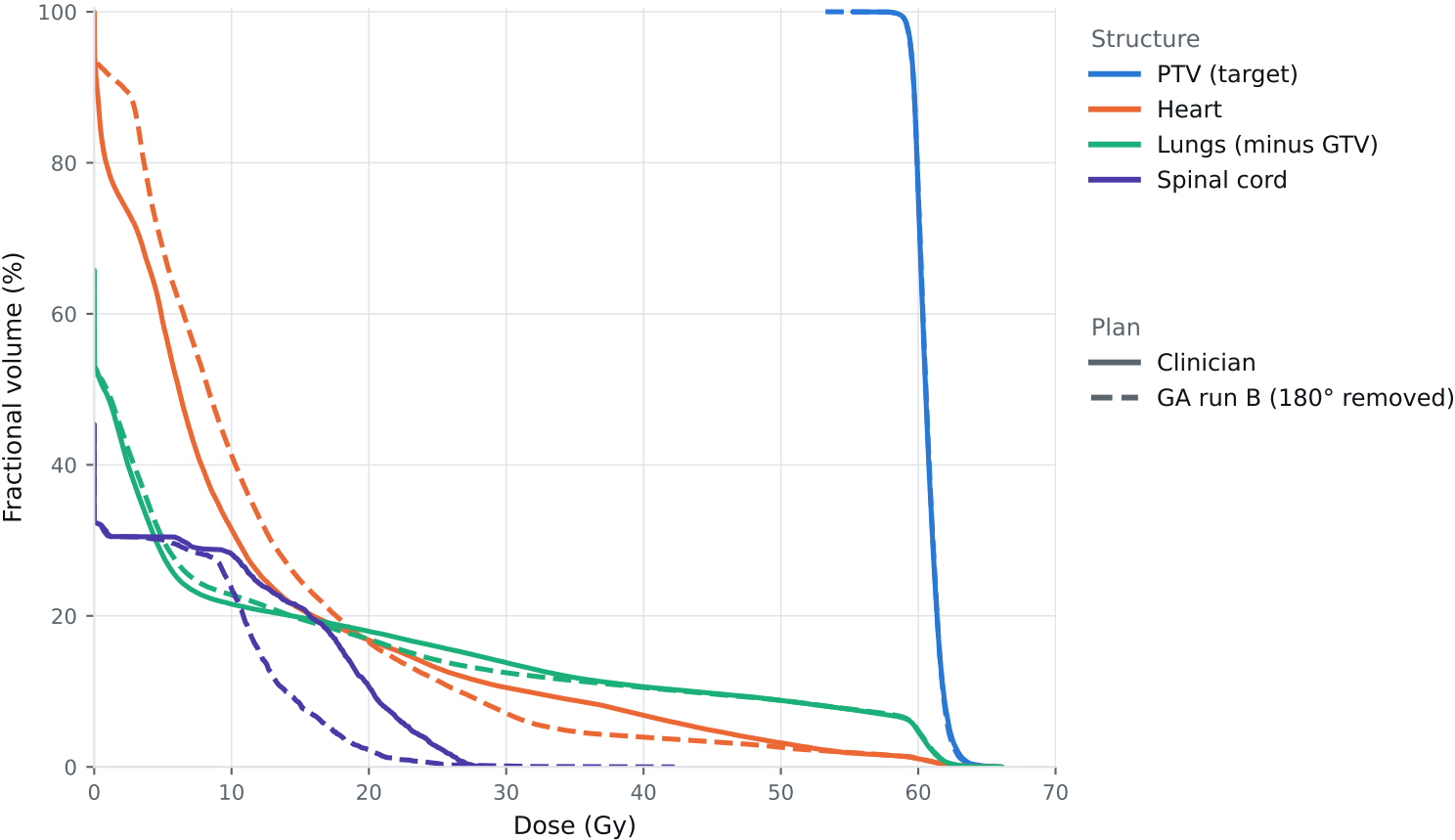


Dose-volume histogram — Lung_Patient_2

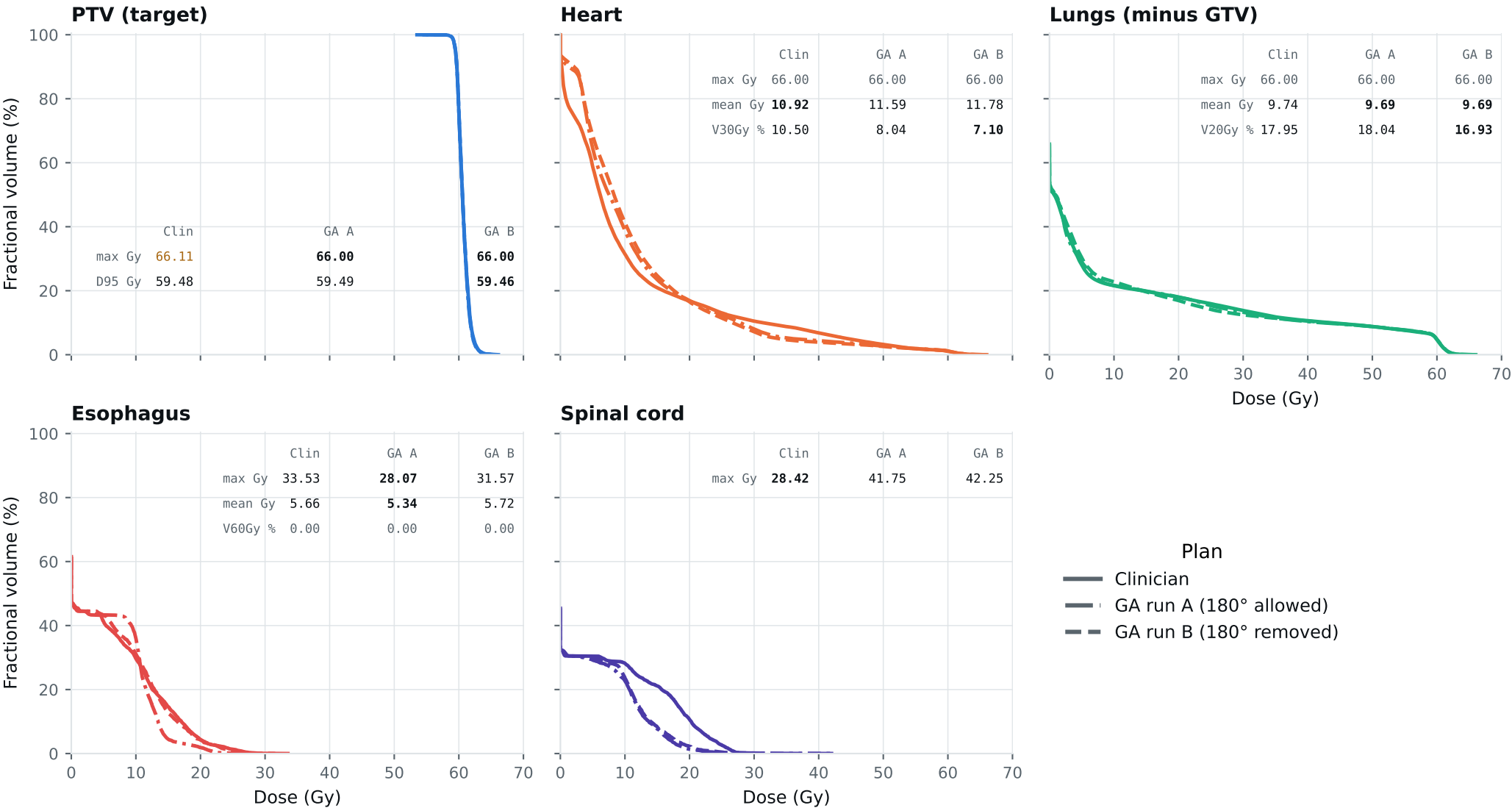
Clinician plan vs. GA run B (180° removed). A curve further left and lower means less dose to that structure.



Curves further left are better for organs; the PTV curve should stay high, then drop sharply. Target curves should overlap; organ curves show the tradeoffs. Esophagus is on page 2.

DVH by structure — Lung_Patient_2

Same curves, one structure per panel, all plans, each annotated with that structure's protocol metrics.
Bold = clinically preferable: organs favor lower dose; PTV favors the value closest to goal within the limit. Grey = all plans are within 0.01 of each other. Amber misses a goal,



Clinical criteria — Lung_Patient_2

Organs: lower is better. PTV: closest to goal without exceeding the limit. Bold marks the best plan.

Criterion	Limit	Goal	Clinician	GA run A 180° allowed	GA run B 180° removed	
GTV max	69	66	65.08	65.25	64.98	Gy
PTV max	69	66	66.11	66.00	66.00	Gy
ESOPHAGUS max	66	—	33.53	28.07	31.57	Gy
ESOPHAGUS mean	34	21	5.66	5.34	5.72	Gy
ESOPHAGUS V60Gy	17	—	0.00	0.00	0.00	%
HEART max	66	—	66.00	66.00	66.00	Gy
HEART mean	27	20	10.92	11.59	11.78	Gy
HEART V30Gy	50	48	10.50	8.04	7.10	%
LUNG_L max	66	—	14.87	33.18	32.13	Gy
LUNG_R max	66	—	66.00	66.00	66.00	Gy
CORD max	50	48	28.42	41.75	42.25	Gy
SKIN max	60	—	52.35	57.35	54.00	Gy
LUNGS_NOT_GTV max	66	—	66.00	66.00	66.00	Gy
LUNGS_NOT_GTV mean	21	20	9.74	9.69	9.69	Gy
LUNGS_NOT_GTV V20Gy	37	—	17.95	18.04	16.93	%
PTV D95 (target coverage)			59.48	59.49	59.46	
Objective value (search signal)			87.55	84.29	85.19	
Solve time			42 s	42 s	43 s	

Beam angles (gantry°)

Clinician0, 185, 210, 240, 270, 300, 330

GA run A30, 165, 180, 195, 255, 285, 330

GA run B45, 165, 195, 255, 285, 300, 330

Grey rows: all plans agree within 0.01 — heart, lung and LUNGS_NOT_GTV max sit at 66 Gy because those structures abut the target. Amber misses the goal; red exceeds the limit.

Source: Lung_Patient_2_clinical_metrics_full_resolution.json, full resolution. The curves and table come from the same solve.